

BRAINIFY

Quiz Application

Focus on Action, Belief and Self-Worth

Project Details

University of Narowal
Subject: Mobile App Development
Platform: Android (Android Studio)
Academic Year: 2023 - 2027

Team Members

Muhammad Talha | Nahah Butt

Table of Contents

Table of Contents.....	2
1. Project Overview.....	2
2. Team Members.....	2
3. Objectives.....	3
4. Key Features.....	3
4.1 Dynamic Quiz Engine	3
4.2 Instant Feedback.....	3
4.3 Score Tracking & History	3
4.4 Interactive Material Design UI	3
4.5 PDF Result Generation	3
5. Technical Details	3
6. Future Work	4
7. Conclusion.....	4
8. References	5

1. Project Overview

Brainify is an advanced Android-based quiz application designed to gamify the learning experience for students. Built using Android Studio, it provides a fast, interactive self-assessment platform that transforms knowledge testing into an engaging challenge rather than a tedious chore.

The application targets students who want to test and strengthen their knowledge across multiple subjects and topics. With its clean Material Design interface and real-time feedback system, Brainify makes learning competitive, fun, and measurable.

2. Team Members

Name	Student ID	Role
Muhammad Talha	23-UON-0446	Lead Developer / Project Manager / Testing, PDF Report & Documentation
Nahah Butt	23-UON-0338	Android UI Developer / Backend Logic and Quiz Engine

3. Objectives

- Make learning fun, interactive, and engaging for students through gamification.
- Improve knowledge retention by providing instant feedback after each quiz attempt.
- Provide a smooth, responsive, and visually appealing mobile experience on Android.
- Allow students to track their performance history and identify weak areas.
- Generate a downloadable PDF result report with quiz details and student information.
- Support multiple subjects, topics, and difficulty levels for diverse use cases.

4. Key Features

4.1 Dynamic Quiz Engine

Questions are served dynamically with randomization across different categories and difficulty levels. This ensures no two quiz sessions feel identical and prevents rote memorization of answer patterns.

4.2 Instant Feedback

After each answer, the app immediately validates the response and highlights whether it was correct or incorrect. This real-time feedback loop accelerates learning and reinforces correct knowledge.

4.3 Score Tracking & History

The application maintains a complete history of past quiz attempts, including scores, topics attempted, and timestamps. Students can review their performance trends over time.

4.4 Interactive Material Design UI

Brainify uses Google's Material Design guidelines to deliver a clean, modern, and intuitive interface. Animations, card layouts, and color-coded feedback make the user experience both attractive and functional.

4.5 PDF Result Generation

Upon completing a quiz, students can export their result as a PDF document. The report includes the student's name, registered email, quiz subject, topic, score, and date — making it shareable with teachers or for personal records.

5. Technical Details

Component	Detail
Platform	Android
Development IDE	Android Studio
Programming Language	Java / Kotlin
UI Framework	Material Design Components
Min Android Version	Android 6.0 (API 23)
PDF Generation	Android PDF API / iText Library
Data Storage	SharedPreferences / SQLite
Architecture	MVC (Model-View-Controller)

6. Future Work

The following enhancements are planned for future versions of Brainify:

- AI-based Personalized Learning — Recommend topics and questions based on individual student performance history using machine learning algorithms.
- Multiplayer Mode — Allow students to compete against each other in real-time quiz battles.
- Adaptive Difficulty Levels — Automatically adjust question difficulty based on the student's current performance in the session.
- Online Quiz Support — Enable teachers to upload and manage quiz content through a web portal, with students accessing quizzes online.
- Gamification System — Introduce badges, rewards, streaks, and leaderboards to increase student motivation and daily engagement.
- Progress Analytics & Reports — Provide detailed graphical reports showing performance trends, topic-wise breakdown, and improvement over time.
- Teacher / Admin Panel — A dedicated dashboard for educators to create quizzes, monitor class performance, and manage students.
- Multi-language Support — Localize the application to support Urdu and other regional languages for broader accessibility.

7. Conclusion

Brainify successfully demonstrates how mobile technology can be leveraged to transform traditional learning into an engaging, gamified experience. By combining a clean Material Design interface with

dynamic quiz generation, real-time feedback, score history, and PDF reporting, the application delivers a complete self-assessment tool for students.

The project gave the team hands-on experience with Android application development, UI/UX design principles, data persistence, and PDF generation on mobile platforms. With a strong foundation already in place, Brainify is well-positioned for future enhancements including AI-driven personalization, multiplayer features, and a full teacher panel — making it a scalable and impactful educational tool.

8. References

- Google LLC. Android Developer Documentation. <https://developer.android.com>
- Material Design Guidelines. <https://m3.material.io>
- Android Studio Official Site. <https://developer.android.com/studio>
- iText Group. iText PDF Library for Android. <https://itextpdf.com>
- University of Narowal — Department of Computer Science, Course Guidelines 2024-2025.